

MATH 4360 HOMEWORK ASSIGNMENT 4

DUE ON THURSDAY 21 SEPTEMBER 2023

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Decide if

$$x^4 - 10x^2 + 1$$

is irreducible over $\mathbb{Q}$.

- (2) Show that

$$x^4 - 4x^3 + x^2 - 4x + 26$$

is irreducible over $\mathbb{Q}$.

- (3) (a) Let $n \in \mathbb{N}$ and a be a square free integer with $|a| \neq 1$. Show that

$$x^n + a$$

is irreducible over $\mathbb{Q}$.

- (b) Show that there exist irreducible polynomials of any degree.

- (4) Let a , b , and c be odd integers. Show that

$$ax^2 + bx + c$$

is irreducible over $\mathbb{Q}$.